MATH INQUIRERS HOLIDAY PROGRAM

Parent Practice Pack

Extra puzzles for every lesson, from warm-up to challenge, with answers at the back

Notice it. Test it. Explain why.

A parent's guide with extra questions for Primary 1

Notice it. Test it. Explain why.

How to help

- Ask before you tell. Start with "What do you notice?" and let your child answer first.
- Use things you have at home: coins, buttons, spoons, pasta, building blocks. Try it with objects before writing.
- If your child is stuck, give hints in three steps: look again, try it with objects, then try a smaller example. Say the answer last.
- Ask "How do you know?" even when the answer is right. The reason matters as much as the answer.
- Praise the thinking: "I like how you checked that." Do not say "you are clever".
- Keep it short. Ten minutes, two or three questions, is enough. Stop while it is still fun.
- Answers are at the back. Please look only after your child has tried.

The levels: Core questions are for everyone. Think Harder questions stretch the idea. Challenge questions are for strong thinkers, and it is fine to leave them for another day.

Lesson 1: Number Patterns

Core

1. Write the next two numbers and the rule: 4, 7, 10, 13, __, __

WORKING

2. Write the next two numbers and the rule: 5, 10, 15, 20, __, __

WORKING

3. Write the next two numbers and the rule: 1, 5, 9, 13, __, __

WORKING

4. Write the next two numbers and the rule: 50, 45, 40, 35, __, __

WORKING

Think harder

5. Fill in the missing numbers: 3, 8, __, 18, __, 28

WORKING

6. Fill in the missing numbers: 40, __, 28, __, 16, 10

WORKING

7. A pattern starts at 3 and adds 5 each time. What is the 8th number?

WORKING

Challenge

8. Frog A jumps in 3s from 0. Frog B jumps in 4s from 0. Which numbers up to 40 do both frogs land on? (Do not count 0.)

WORKING

9. I am in the counting patterns of 3s and of 5s. I am more than 10 and less than 40. Which numbers can I be?

WORKING

Explain why: How do you know your answer is right?

Lesson 2: Number Bonds

Core

1. 10 = 3 + __

WORKING

2. 10 = __ + 8

WORKING

3. 10 = 6 + __

WORKING

4. 20 = 5 + __

WORKING

Think harder

5. 5 + 4 = 6 + __

WORKING

6. 8 + 2 = 7 + __

WORKING

7. 9 + 6 = __ + 10

WORKING

Challenge

8. Put 1, 2, 3, 4, 5 and 6 in the triangle so each side adds to 9. Corners 1, 2 and 3. What goes in the middle of each side?

WORKING

9. Corners 4, 5 and 6, and the numbers 1, 2, 3 in the middles. What do all the sides add to? Where does each middle number go?

WORKING

Explain why: How do you know your answer is right?

Lesson 3: Picture Graphs

Core

1. A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the most votes?

WORKING

2. A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the fewest votes?

WORKING

3. A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many more children chose orange than grape?

WORKING

4. A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many children voted altogether?

WORKING

Think harder

5. 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children chose blue?

WORKING

6. 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children voted altogether?

WORKING

7. 1 picture = 2 children. Red has 3 pictures and blue has 5. How many more children chose blue than red?

WORKING

Challenge

8. 18 children voted. 7 chose apple and 5 chose orange. The rest chose grapes. How many chose grapes?

WORKING

9 Votes: A 8, B 5, C 6. Then 4 more children join and all choose B. Which fruit wins now, and by how many votes?

WORKING

Explain why: How do you know your answer is right?

Lesson 4: Repeating and Growing Patterns

Core

1. The pattern is circle, square, triangle, circle, square, triangle, circle … What is the 10th one?

WORKING

2. The pattern is star, heart, star, heart, star, heart … What is the 9th one?

WORKING

3. The pattern is red, red, blue, red, red, blue, red … What is the 8th one?

WORKING

4. The pattern is A, B, C, D, A, B, C … What is the 11th one?

WORKING

Think harder

5. Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square … What is the 14th shape?

WORKING

6. Same pattern: how many triangles are in the first 20 shapes?

WORKING

7. Same pattern: in which places do the squares come? Write the first five.

WORKING

Challenge

8. Squares in a row use 4 sticks for 1 square, 7 sticks for 2 squares and 10 sticks for 3 squares. How many sticks for 8 squares?

WORKING

9. Using the same pattern, how many squares can you make with 22 sticks?

WORKING

Explain why: How do you know your answer is right?

Lesson 5: Working Backwards

Core

1. I think of a number. I add 3. I get 8. What is my number?

WORKING

2. I think of a number. I add 6. I get 14. What is my number?

WORKING

3. I think of a number. I add 5. I get 12. What is my number?

WORKING

4. I think of a number. I add 9. I get 17. What is my number?

WORKING

Think harder

5. I think of a number. I add 5, then I take away 3. I get 9. What is my number?

WORKING

6. I think of a number. I add 6, then I take away 2. I get 10. What is my number?

WORKING

7. I think of a number. I add 4, then I take away 7. I get 8. What is my number?

WORKING

Challenge

8. I think of a number. I add 6, take away 4, then add 5. I get 17. What is my number?

WORKING

9. A number machine has two steps: add 3, then add 4. IN 5 gives OUT 12. What is IN when OUT is 20?

WORKING

Explain why: How do you know your answer is right?

Lesson 6: Money

Core

1. Pay exactly 15¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

WORKING

2. Pay exactly 25¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

WORKING

3. Pay exactly 30¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

WORKING

4. Pay exactly 40¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

WORKING

Think harder

5. How many ways can you make 25¢ using 20¢, 10¢ and 5¢ coins?

WORKING

6. How many ways can you make 30¢ using 20¢, 10¢ and 5¢ coins?

WORKING

7. How many ways can you make 20¢ using only 10¢ and 5¢ coins?

WORKING

Challenge

8. Make 35¢ using exactly 3 coins from 5¢, 10¢, 20¢ and 50¢. How many ways are there?

WORKING

9. Make 40¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. Find them all.

WORKING

Explain why: How do you know your answer is right?

Lesson 7: 2D Shapes

Core

1. How many sides and corners does a triangle have?

WORKING

2. How many sides and corners does a square have?

WORKING

3. How many sides and corners does a pentagon have?

WORKING

4. How many sides and corners does a hexagon have?

WORKING

Think harder

5. I have 4 sides. All my sides are the same length. What shape am I?

WORKING

6. How many sides do 3 triangles have altogether?

WORKING

7. How many corners do 5 squares have altogether?

WORKING

Challenge

8. A square has both diagonals drawn. How many triangles can you find?

WORKING

9. A big triangle has 1 line from the top corner to the bottom. How many triangles? What about 2 lines? 3 lines?

WORKING

Explain why: How do you know your answer is right?

Lesson 8: Logic and Sorting

Core

1. Ali is taller than Ben. Ben is taller than Mei. Who is the tallest? Who is the shortest?

WORKING

2. The cat is heavier than the hen. The hen is heavier than the mouse. Who is the lightest?

WORKING

3. Red, blue and green ribbons. Red is longer than blue. Green is shorter than blue. Which ribbon is the longest?

WORKING

4. Which number is the odd one out: 2, 4, 6, 9, 10? Why?

WORKING

Think harder

5. In a race, Sam is ahead of Tia. Tia is ahead of Raj. Mei is ahead of Sam. Who is third?

WORKING

6. Tia does not drink milk. Sam drinks neither milk nor juice. Raj drinks the third drink. The drinks are milk, juice and water. Who drinks what?

WORKING

7. Numbers 1 to 12. Which are multiples of 2? Which are multiples of 3? Which are both?

WORKING

Challenge

8. Four friends stand in a line. Ali is not first. Ben is ahead of Ali. Sam is last. Mei is behind Ali. In what order do they stand?

WORKING

9. Three boxes are labelled “Apples”, “Oranges” and “Mixed”. Every label is wrong. You may take one fruit from one box. Which box should you pick from, and why?

WORKING

Explain why: How do you know your answer is right?

Lesson 9: Length

Core

1. A pencil is 7 cubes long. A pen is 10 cubes long. How many cubes longer is the pen?

WORKING

2. A ribbon is 12 cubes long. You cut off 5 cubes. How long is the piece left?

WORKING

3. Ali's book is 9 cubes long and Mei's book is 6 cubes long. Whose is longer? By how much?

WORKING

4. Which is longer: 8 cubes or 8 paper clips? Why can we not tell?

WORKING

Think harder

5. 1 clip is as long as 3 cubes. A pen is 12 cubes long. How many clips long is the pen?

WORKING

6. 1 clip is as long as 3 cubes. A ruler is 5 clips long. How many cubes long is the ruler?

WORKING

7. Ali's desk is 6 spans long. Mei's desk is 9 spans long, but they measured the same desk. Whose hand is bigger?

WORKING

Challenge

8. A rope is 20 cubes long. Ali cuts it into pieces that are each 4 cubes long. How many pieces does he get? How many cuts does he make?

WORKING

9. A train has 3 carriages. Each is 6 cubes long. There is a gap of 1 cube between carriages. How long is the whole train?

WORKING

Explain why: How do you know your answer is right?

Lesson 10: Mixed Puzzles

Core

1. Count in 3s: 3, 6, 9, __, __

WORKING

2. 10 = 4 + __ and 20 = 12 + __

WORKING

3. Red 4, blue 6, green 3. How many more are blue than green?

WORKING

4. I think of a number. I add 4. I get 9. What is it?

WORKING

Think harder

5. A pattern starts at 5 and adds 3 each time. What is the 10th number?

WORKING

6. Ali has 12 stickers. He gives 5 to Mei and then gets 3 more. How many does he have now?

WORKING

7. Which three coins from 5¢, 10¢, 20¢ and 50¢ make 45¢?

WORKING

Challenge

8. Two numbers add to 12 and differ by 2. What are they?

WORKING

9. 1 triangle uses 3 sticks, 2 triangles use 5 sticks. How many sticks for 9 triangles? How many triangles with 25 sticks?

WORKING

Explain why: How do you know your answer is right?

Answers (look after your child has tried)

Lesson 1: Number Patterns

1. 16, 19. Add 3.

2. 25, 30. Add 5.

3. 17, 21. Add 4.

4. 30, 25. Take away 5.

5. 13 and 23. Add 5.

6. 34 and 22. Take away 6.

7. 38

8. 12, 24 and 36.

9. 15 and 30.

Lesson 2: Number Bonds

1. 7

2. 2

3. 4

4. 15

5. 3

6. 3

7. 5

8. Between 1 and 2: 6. Between 2 and 3: 4. Between 3 and 1: 5.

9. Each side adds to 12. Between 4 and 5: 3. Between 5 and 6: 1. Between 6 and 4: 2.

Lesson 3: Picture Graphs

1. Orange

2. Grape

3. 5 (9 – 4)

4. 25 (7 + 5 + 9 + 4)

5. 10

6. 24 (12 pictures × 2)

7. 4 (2 more pictures × 2)

8. 6

9. B wins with 9, which is 1 more than A.

Lesson 4: Repeating and Growing Patterns

1. circle

2. star

3. red

4. C

5. triangle

6. 10 (5 groups of 4, with 2 triangles in each group)

7. 4, 8, 12, 16, 20

8. 25 (4 + 3 × 7)

9. 7 (22 – 1 = 21, 21 ÷ 3 = 7)

Lesson 5: Working Backwards

1. 5

2. 8

3. 7

4. 8

5. 7

6. 6

7. 11

8. 10 (17 – 5 + 4 – 6)

9. 13 (the same as adding 7)

Lesson 6: Money

1. 5¢ + 10¢

2. 5¢ + 20¢

3. 10¢ + 20¢

4. 20¢ + 20¢

5. 4 ways. (20+5; 10+10+5; 10+5+5+5; 5×5)

6. 6 ways.

7. 3 ways. (10+10; 10+5+5; 5×4)

8. 1 way: 20¢ + 10¢ + 5¢

9. 2 ways: 20¢ + 10¢ + 5¢ + 5¢; 10¢ + 10¢ + 10¢ + 10¢

Lesson 7: 2D Shapes

1. 3 sides and 3 corners

2. 4 sides and 4 corners

3. 5 sides and 5 corners

4. 6 sides and 6 corners

5. A square

6. 9

7. 20

8. 8 (4 small and 4 made of two small ones)

9. 3, 6 and 10 triangles.

Lesson 8: Logic and Sorting

1. Ali is tallest; Mei is shortest.

2. The mouse

3. Red

4. 9. It is the only odd number.

5. Tia (order: Mei, Sam, Tia, Raj).

6. Sam: water. Tia: juice. Raj: milk.

7. Multiples of 2: 2, 4, 6, 8, 10, 12. Multiples of 3: 3, 6, 9, 12. Both: 6 and 12.

8. Ben, Ali, Mei, Sam.

9. Pick from the box labelled “Mixed”. Its label is wrong, so it holds only one kind of fruit. If you take out an apple, that box is Apples. The box labelled “Oranges” cannot be oranges, and apples are taken, so it is Mixed. The box labelled “Apples” must be Oranges.

Lesson 9: Length

1. 3

2. 7 cubes

3. Ali’s, by 3 cubes.

4. We cannot tell. The units are different sizes.

5. 4 clips

6. 15 cubes

7. Ali’s. A bigger span needs fewer spans to cover the same desk.

8. 5 pieces and 4 cuts.

9. 20 cubes (18 + 2)

Lesson 10: Mixed Puzzles

1. 12, 15

2. 6 and 8

3. 3

4. 5

5. 32

6. 10

7. 20¢ + 20¢ + 5¢

8. 5 and 7

9. 19 sticks for 9; 12 triangles for 25 sticks.